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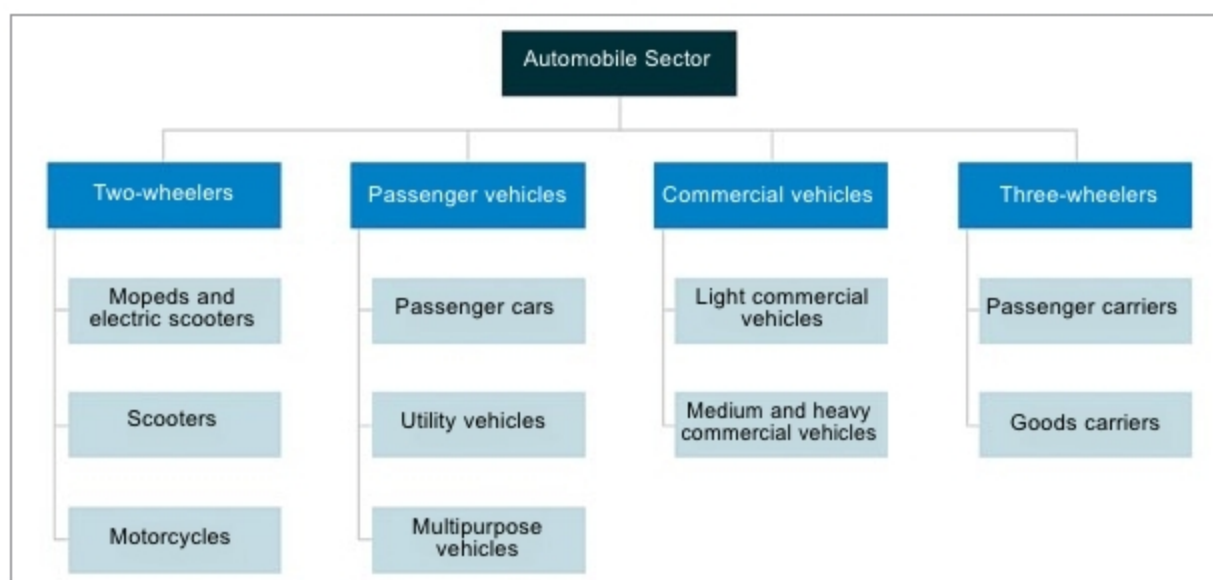


Fig. 2: The Indian automotive market segments (Source: IBEF)

conductors and so on.

Key electronic components used in EVs include microcontrollers (MCUs), rectifiers, analogue gate drivers, EEPROM memories, protection devices, voltage regulators, power MOSFETs, IGBTs, and ICs for power management and battery monitoring systems.

Moreover, complete solutions for electric traction and energy recovery, power steering, automated manual transmission, cooling system, pumps and power management are also in demand. This, in turn, enhances the opportunity to increase the overall electronics content of the cars. Innovative automotive applications like intelligent power switches for anti-lock brake systems, micro-electromechanical systems (MEMS) inertial sensors for automotive airbags and telematics microprocessors also see demand.

The EV market will also help drive the growth of IGBTs, which have long been at home in the switch power supply market and comprise the largest segment of the market for EV power systems. Vehicle electrification involves voltage levels between 12 and 400 volts, and requires high-power IGBT modules for drive and traction applications.

Another area of application is wheel management. Wheels may become electrical, generating electricity, and hybrids may have thermoelectric on exhaust systems and turbines in them, both generating useful kilowatt-level power. Thus, wheel management will also require electronics.

Desire for longer driving between charging points, faster battery charging times, increasing demand for connectivity and infotainment, safety and security

may drive the use of electronics beyond power or wheel management. For example, need for electrical active suspension leading to better ride and fuel economy will involve more electronic components and devices. Car entertainment systems with near-field communications (NFC) technology, car access and in-vehicle networks, and all features related to secure connected mobility will enhance electronics content.

Moving forward

Industry 4.0—enabling innovative processes, digital manufacturing and automation—has the potential to disrupt the industrial value chain and push automotive sector to rethink the way they do business. These shifts could also modify typical job profiles required by auto component manufacturers. For example, instead of a manual worker in the production line, companies might need an exception handler or an operator in an automated workflow. The role of a supply chain planner could evolve from merely planning the logistics of supply in a segmented approach to owning a fully-integrated supply chain, from order to delivery.

Moreover, stringent regulations are evolving so rapidly that the automotive sector might need to rethink about their operational aspects to comply with regulatory and other guidelines for the following:

Emissions. Enforcement of BS-VI norms by 2020 push towards EVs, discussions on methanol, CNG and potentially even fuel cells.

Safety. Mandatory ABS on cars, buses and two-wheelers, cabin safety, and roll-over protection systems. **EFY**